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Pairing Caregivers as a Preliminary Step in the Home Healthcare Routing and Scheduling Problem

by

Nolan Willoughby

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Dr. Jörg Kalcsics

Abstract

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Own Work Declaration

I attest that I am the sole author of this thesis and that all the content within it is my original work, except for any instances where I have clearly indicated otherwise in the text.

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1 Introduction

Section 2 will provide a review of relevant literature to the project in the interest of contextualizing the general nature of the work being done. Section 3 will formally define the problem being addressed by this work and the specific context of one home healthcare company. Section 4 will cover all approaches considered as potential means of addressing the problem, the specific approach chosen, and the way in which that approach was implemented, while Section 5 will provide detailed computational results of the methodology as implemented. Section 6 will conclude by briefly reviewing the accomplishments detailed throughout, detailing limitations of the work, and noting recommendations for future development of the work.

2 Background

To better inform the work conducted, it is necessary to first understand the literature context in which it is placed. This chapter will provide such context by reviewing relevant literature on Home Healthcare as a service and industry, Operational Research and its relevant canon problem structures, and applications of Operational Research to this and other healthcare contexts.

2.1 Home Healthcare

Home Healthcare is the process by which certain health-related services are provided to patients in their home. These services typically involve nursing care, though can also consist of prescription and other medical supply delivery as well as assistance with day-to-day activity [1]. These services provided by home healthcare models are becoming increasingly essential as demographic changes yield older populations, greater geographical dispersion relative to both family and central hospital campuses, and a general preference toward aging in place. Additionally, home healthcare services can prove essential to modern healthcare systems by enabling personalized, expert care for patients with chronic illness, mobility challenges, and post-op recovery needs who may not otherwise be able to receive such quality care [2, 5].

Consequently, the importance of home healthcare companies, i.e. companies providing these services, is of increasing social interest. These groups must meet several operational challenges to meet the needs of their client base, and must do so with efficiency to ensure competitive advantage in a market system. Among the challenges of these groups is the need to find ways to efficiently provide diverse services requiring differing numbers and types of caregivers [3, 5]. Not only must these services be provided by varying numbers of caregivers with various qualification, they must also be delivered within strict time windows according to patient care needs [3]. Given the operational complexity and social importance of these operations, the operations of home healthcare companies are a domain with high-potential for operations research applications.

2.2 Operational Research and its Relevance

Operational Research is the branch of applied mathematics dealing with the formulation and solving of decision problems. As such, it presents an area of particularly enticing growth potential for operations involving multiple, complex decisions. As discussed in a prior section, the field of home healthcare is one such domain, as it involves decisions on the routing of visits, the scheduling of visits, and the assignment of caregivers to those visits.

The first relevant problem structure from the OR canon to note is the Vehicle Routing Problem. Formally, the Vehicle Routing Problem is defined by a set of customers, each with known demand, a set of vehicles available to serve those customers, and known costs for serving each customer both individually and following from a prior customer. The objective of the Vehicle Routing Problem is to then assign each vehicle a subset of the customers and an order, i.e. routing, by which each vehicle is to serve its assigned customers such that the overall cost across all vehicles' assigned routes is minimized [7]. This problem typically involves capacitated vehicles, i.e. each vehicle is limited in the amount of demand it is able to service, while otherwise resembling an extended version of the Travelling Salesman Problem.

In particular, there exists a canon structure of the Vehicle Routing Problem with Time Windows. This structure extends the existing Vehicle Routing Problem structure to enforce time windows within which a visit must be scheduled[8].

The next relevant problem structure to consider is the Assignment Problem.

A final problem structure to consider is the Synchronization Problem.

2.3 Application of OR to Healthcare Contexts

2.3.1 The Home Healthcare Routing and Scheduling Problem

2.3.2 Special Considerations

Due to the significant human impact of healthcare contexts, certain considerations ought to be made to ensure positive, equitable outcomes when applying OR to these contexts.

3 Problem Statement

As a first step in an engineering design process, it is important to formally define the nature and objective of this work. This case involves the scheduling and routing decisions faced by a home healthcare company. The company employs approximately 50 caregivers and services approximately 150 clients. Caregivers may work either a full-time, morning, or afternoon shift pattern, and each caregiver is available for a given subset of days in the planning window. This information is known and decided on a two week planning horizon due to minimum work thresholds set by the UK Home Office for caregivers on sponsored work visas. Caregivers may or may not have the ability to drive, with driving caregivers able to cover a larger service area than non-drivers. That said, caregivers are generally not compensated for time spent on their shift travelling, though there can be exceptions to this rule for certain, extenuating circumstances. Additionally, approximately 90% of all caregivers identify as female, and all caregivers share functionally equivalent qualifications sufficient to perform any visit the company may have committment to perform.

Clients typically require between three and five visits daily, with most visits lasting between 30 and 60 minutes. Each of these visits must occur within a given time window, and each visit requires either one or two caregivers present. Clients typically prefer visits to be performed by caregivers already known to the client, and clients can specify a preference on the gender of the caregiver(s) assigned to perform their visit(s). Clients and their caregivers are spread across the company's service area, which covers three municipalities. The distribution of clients and caregivers across this area is not uniform, and this disparity must be taken into account when routing and scheduling visits. On any given day, the total duration of all visit minutes requiring two caregivers present represents about 40% of the total duration of all visit minutes which must be served on that day.

The problem in this case, therefore, is to identify an optimal or near-optimal allocation or scheduling of caregivers to the set of visits the company is committed to service and the order or routing in which the caregivers ought to perform their assigned visits. Attention must be paid to the shift pattern that assigned caregivers work, whether assigned caregivers are able to drive or are paired with another caregiver able to drive, and whether caregivers match the gender preferences of the client. Because all caregivers functionally share the same qualifications, there is no need in this case to consider visits that can only be performed by a subset of caregivers, with the exception of cases with an expressed gender preference. This is in contrast to most formulations of the Home Healthcare Routing and Scheduling Problem, which constrain a caregiver's scheduling according to whether that caregiver has the requisite qualification(s) to perform the tasks of each visit. With the problem formally defined, it is then possible to lay forth a modelling approach meant to solve the problem as defined.

4 Methodology

Having now defined the problem, this chapter will detail the methodology considered and ultimately chosen to address the problem. Discussion will start with a review of the various approaches considered as potentially applicable to the problem, followed by an overview of the approach deemed most appropriate to the context. The rest of the chapter will then focus on the specifics of the ultimate design approach taken.

4.1 Design Considerations

Company wants to service its clients to their satisfaction. Caregivers want good routing and scheduling that increases their share of working hours which are paid. Clients want good, timely visits from caregivers they know and trust. Potential objectives therefore include. The least computationally complex model would choose exactly one objective, while there exists computational trade-off between computational complexity and solutions improved across all objectives.

Some visits require multiple caregivers. Synchronization problem is a thing applicable to this in certain companies' contexts. Synchronization approach yields addition of operational and computational complexity and lack of schedule/routing robustness in the event of unanticipated disruptions. Pairing caregivers for the day may lead to routings and schedulings marginally less efficient than a synchronization approach may yield, but would be less operationally complex, less operationally vulnerable, and ultimately likely less computationally complex than a synchronization approach.

There is also the question of considering gender preference, and when in the decision pipline that consideration should be made.

4.2 Modelling Approach

4.3 Formulation

Sets

D	set of days to schedule caregivers
C^d	set of caregivers available on day d
U^d	set of feasible caregiving units available on day d
$U_p^d \subset U^d$	set of feasible caregiving units of effective size two available on day d
$U_s^d \subset U^d$	set of feasible caregiving units of effective size one available on day d
$U_D^d \subset U^d$	set of feasible caregibing units available on day d containing an equivalent full-time driver
L	set of localities served

Parameters

d_{ij}	driving time between caregivers $i \in C^d$ and $j \in C^d$
d_u	driving time required to create unit u
r_{ul}	driving time required for unit u to reach locality l
F_u^d	total duration of pair visits on day d where both active caregivers in unit $u \in U_p^d$ are familiar to the client
f_u^d	total duration of pair visits on day d where either active caregiver in unit $u \in U_p^d$ is familiar to the client
s_u^d	total duration of solo visits on day d where the active caregiver in unit $u \in U_s^d$ is familiar to the client
V^d	total duration of all visits on day d
V_P^d	total duration of all visits on day d requiring two caregivers present
V_s^d	total duration of all visits on day d requiring one caregiver present
$p^d = \frac{V_P^d}{V^d}$	share of total visit duration on day d requiring a caregiving pair
$T^d = \frac{p\lvert C_D^d \rvert}{1+p}$	target number of caregiving pairs to assign on day d
V_{Pl}^d	total visit duration in locality l on day d requiring a caregiving pair
V_{sl}^d	total visit duration in locality l on day d requiring a single caregiver
l_u	binary indicator whether caregiver unit u is located in locality l
K	maximum distance carers will travel for pairing

Decision Variables

$$x_u^d = \begin{cases} 1 & \text{if caregiving unit } u \in U^d \text{ is assigned on day } d \\ 0 & \text{else} \end{cases}$$

$$z_{ul}^d = \begin{cases} 1 & \text{if caregiving unit } u \in U_D^d \text{ is assigned to locality } l \in L \text{ on day } d \\ 0 & \text{else} \end{cases}$$

w_{ul}^d variable representing the linearization of the product of binary variables $z_{ul}^d x_u^d$

$$
\begin{aligned}
\max \quad & \sum_{d\in D}\left(\sum_{u\in U_s^d} s_u^d x_u^d + \sum_{v\in U_p^d}(F_v^d + f_v^d)x_v^d\right) && && (4.1)\\
\min \quad & \sum_{d\in D}\left(\sum_{u\in U^d}\left(d_u x_u^d + \sum_{l\in L} r_{ul} w_{ul}^d\right)\right) && && (4.2)\\
\text{s.t.} \quad & \sum_{u\in\{u\in U^d \mid i\in u\}} x_u^d = 1 && \forall i\in C^d,\ d\in D && (4.3)\\
& \sum_{u\in U_p^d} w_{ul}^d \le \lceil \tfrac{V_{Pl}^d}{V_P^d} T^d \rceil + \epsilon && \forall d\in D,\ l\in L && (4.4)\\
& \sum_{u\in U_p^d} w_{ul}^d \ge \lfloor \tfrac{V_{Pl}^d}{V_P^d} T^d \rfloor - \epsilon && \forall d\in D,\ l\in L && (4.5)\\
& \sum_{u\in U_s^d} w_{ul}^d \le \lceil |C^d| - T\epsilon \rceil + 2\epsilon && \forall d\in D,\ l\in L && (4.6)\\
& \sum_{u\in U_s^d} w_{ul}^d \ge \lfloor |C^d| - T \rfloor - 2\epsilon && \forall d\in D,\ l\in L && (4.7)\\
& T - \epsilon \le \sum_{u\in U_p^d} x_u^d \le T + \epsilon && \forall d\in D && (4.8)\\
& |C^d| - T - 2\epsilon \le \sum_{u\in U_s^d} x_u^d \le |C^d| - T + 2\epsilon && \forall d\in D && (4.9)\\
& x_u^d = 0 && && (4.10)\\
& && \forall u\in\{\, u\in U^d \mid d_u \ge 2K \lor d_{ij} \ge K \forall (i,j)\in u \,\} && \\
& 0 \le w_{ul}^d \le x_u^d && \forall u\in U_D^d,\ l\in L,\ d\in D && (4.11)\\
& w_{ul}^d \le z_{ul}^d && \forall u\in U_D^d,\ l\in L,\ d\in D && (4.12)\\
& w_{ul}^d \ge z_{ul}^d + x_u^d - 1 && \forall u\in U_D^d,\ l\in L,\ d\in D && (4.13)\\
& x_u^d \in \{0,1\} && \forall u\in U^d,\ d\in D && (4.14)\\
& z_{ul}^d \in \{0,1\} && \forall u\in U^d,\ l\in L,\ d\in D && (4.15)\\
& w_{ul}^d \in \mathbb{R} && && (4.16)
\end{aligned}
$$

5 Results

The model was implemented with Gurobi's optimization solver using its Python library.

5.1 Effect on Downstream Routing and Scheduling Process

6 Conclusion

lol we've got a long way to go before i start doing this

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Appendices

A Appendix A